CHAPTER SEVEN

Parts, Motion, and the Development of the Animate-Inanimate Distinction in Infancy

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Taste your legs, sir; put them to motion.

Shakespeare, Twelfth Night

Since things in motion sooner catch the eye than what not stirs.

Shakespeare, Troilus and Cressida

THERE IS NO QUESTION that as adults we are aware of the various properties, abilities, and characteristics of the things around us.[1] In addition to being able to recognize, for example, animals and people on the basis of their external perceptual appearance—that which "can be seen" (Keil, 1989)—we also know that they are biological entities capable of self-propelled motion, and that they possess mental states such as goals, beliefs, and desires. Likewise, we can recognize rocks, plants, toys, and vehicles by their perceptual properties, but we are also aware that the members of these categories engage in a more limited kind of movement—they are, for instance, generally caused to move by other objects—and they are incapable of possessing any kind of psychological states.

This kind of understanding about the physical, biological, and psychological properties of animate entities and inanimate objects is thought by many to be well developed by the time that children reach preschool age (e.g., Gelman & Markman, 1986, 1987; Keil, 1991; Mandler, 1992). For example, there is evidence that 3- and 4-year-olds know whether different ontological kinds—such as a statue, a man, and a marmoset—are able to go up and down a hill "all by itself" (Massey & Gelman, 1988). Likewise, children in this age range show an understanding of several aspects of

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growth and illness (e.g., Backscheider, Shatz, & Gelman, 1993; Springer & Keil, 1991), and they make inductions about category membership on the basis of nonobservable biological properties rather than perceptual appearances, when given a choice between the two (Gelman & Coley, 1990; Gelman & Markman, 1986, 1987). By comparison, very young infants seem oblivious to these deeper, *nonobservable* properties. Indeed, it is now well established that categorization in the first 6 months of life or so is based predominantly on attributes that are readily available in the perceptual array (see e.g., Behl-Chadha, 1996; Quinn & Eimas, 1996b, 1997). It has been shown, for example, that 3- to 4-month-olds in a visual familiarization procedure can form categorical representations of cats that exclude dogs (Quinn, Eimas, & Rosenkrantz, 1993) and of mammals that exclude furniture (Behl-Chadha, 1996; see also Younger & Fearing, 1999). Infants at this age show little understanding, however, of the nonobvious properties commonly associated with animates or inanimates (Rakison & Poulin-Dubois, 2001).

From Perception to Conception?

Although the available evidence points to the fact that young infants categorize objects on the basis of perceptual information—what is called perceptual categorization—and that preschoolers have acquired knowledge and make use of nonobvious information, what is often called conceptual categorization, the process underlying the acquisition of an understanding of the properties that differentiate animate entities from inanimates objects remains somewhat opaque. For many developmentalists, this issue is best portrayed as the "perceptual to conceptual shift" whereby early categories are formed on the basis of perceptual resemblance, whereas later categories, following a stagelike transformation in representational structure and content, are formed primarily on the basis of nonobvious properties (e.g., Gelman & Markman, 1986, 1987; Keil, 1981, 1991; Murphy & Medin, 1985; Piaget, 1952). This dichotomous characterization of early categorization has, however, recently been criticized not only as an oversimplification but also as a mischaracterization of representational change (e.g., Jones & L. Smith, 1993; Madole & Oakes, 1999; Quinn & Eimas, 1997; for a more general criticism of stage-based theories of development, see Siegler, 1996, 2000). For instance, one group of researchers has proposed that no transition or shift occurs because categorization continues to rely on perceptual information throughout the lifespan (Eimas, 1994; Jones & Smith, 1993; Quinn & Eimas, 1997; Rakison & Poulin-Dubois, 2001; Smith & Heise, 1992), whereas others have argued that there is no need for a shift because perceptual and conceptual categorization develop side by side (e.g., Mandler, 2000).

An additional problem that arises in consideration of the development of early knowledge about physical, biological, and psychological properties of objects concerns the ontological status of different attributes. In general, the term *perceptual* tends to refer to that which is readily available in the perceptual input—for example, shape, color, texture, external attributes, and so on—whereas the terms *nonobvious* or *conceptual* tend to refer to internal parts or structures or to specific physically,

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psychologically, or biologically functional properties; for example, possessing blood or a brain, possessing a motor engine, or the ability to fly. Unfortunately, however, the different properties of objects cannot be classified quite as neatly as these characterizations suggest. It is ambiguous, for example, whether properties such as "chases sticks" or "barks" should be thought of as perceptual or conceptual attributes of dogs (Mandler, 1997), and it is unclear where to draw the line between properties that are readily available in the perceptual input—color and shape, for example—and those that are only intermittently available to the perceptual system, such as "gives birth to puppies" or "engages in self-propelled motion."

Theses of the Chapter

It is evident from the discussion thus far that I believe that researchers who focus on early category and concept development for objects in the world need to focus not only on the nature of conceptual structure and its contents but also on the processes of conceptual change whereby various information is integrated into infants' emerging representations for people, animals, and things. In this chapter, I will attempt to examine both of these issues. In particular, I will argue that infants develop representations for objects in the world, and the properties associated with the member of each of those domains, through a process of continual, incremental learning. This view of representational enrichment, which is in many ways akin to that forwarded by Quinn and Eimas (1997, 2000; Eimas, 1994), has at its core the idea that the information that underlies the developing concepts of infants and children is fundamentally perceptual in nature. Thus, one theme of the chapter is that no "perceptual-to-conceptual" shift occurs in infants or young children. To be sure, infants begin to incorporate into their representations information about objects that is considered by many to be nonobvious; yet I will argue that the same associative learning process is applied both for "perceptual" and "nonobvious" information because there is in fact no qualitative difference—both in terms of the input and the representations that result—between them.

A second thesis of the chapter is that infants' representations, and indeed those of older children and adults, are grounded in clusters of correlated attributes. This idea is now over 20 years old (Rosch, Mervis, Gray, Johnson, & Boyes-Braem, 1976; Rosch, 1978) and has resurfaced a number of times in the intervening period as a potential organizing principle of categories and the concepts that characterize them. For instance, Smith and Heise (1992) suggested that through experience infants start to attend to the attributes that are part of correlations, and Madole and Oakes (1999) similarly argued that infants ignore arbitrary form–function correlations because their experience allows them to focus only on meaningful relationships, such as, for example, those in which form predicts function.

These preceding formulations, and those that rely on similarity as the foundation of representations' structure and content, have generally been undermined by an *insufficiency of constraints* argument (e.g., Keil, 1981; Murphy & Medin, 1985). The crux of this argument is that because there are so many correlations available in the

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world, it is impossible to know a priori which ones are significant for category membership and which ones are not. I will argue, however, that infants possess inherent attention biases that, coupled with an associative learning mechanism, direct them to attend to and to encode specific correlations among attributes. These correlations are often those that involve dynamic attributes and that could be considered causally or functionally relevant. In contrast to previous arguments that lay theories bind or structure concepts by making sense of such relationships (Murphy & Medin, 1985), I will argue that feature correlations are grounded in causal relations observed during infancy and early childhood and that theories about feature correlations emerge only when such information is required.

I will start by outlining the three attention biases that, among others, are crucial in directing the infant toward specific kinds of information; namely, those that involve causal relations between static and dynamic attributes. I will then outline in more detail how attention to these relations, in conjunction with an associative learning mechanism, leads to the development of representations that include both perceptual and nonobvious information. I will cite studies from my lab, and from other infancy researchers, to support this view. Finally, I will present data from the inductive generalization paradigm that suggest that infants in the second year of life are only starting to develop associations between different categories of objects and the motion characteristics of those objects.

Early Attention Biases

It has been well documented that, as far as infants are concerned, not all properties and characteristics of things in the world are equally salient. It would be impossible for any species to "carve nature at its joints" if those joints did not in some way stand out. Ontological kinds have properties in common that not only allow them to be seen as similar to one another but also as different from other classes of objects. In this section, I propose a number of inherent attention biases—though other terms such as *constraints* or *learning enablers* could just as easily describe them—that facilitate attention to these properties and in turn allow the development of knowledge about the various categories of things that exist in the world. These biases relate to infants' predilection to attend to parts, the attention-capturing role of movement, and the importance of relative size; however, here I will focus primarily on the effects of motion and object attributes in the development of representations of objects.

It is important to note that I do not hold that these are the only attentional biases that infants possess: It has been shown for example, that neonates orient to faces more than to nonfaces (Morton & Johnson, 1991) and that older infants look longer at complex stimuli than at simple stimuli (Fantz, 1961) and orient their heads to locate a sound (Clarkson, Clifton, & Morrongiello, 1985). Such biases are clearly also important in directing infants' attention toward significant stimuli. However, my claim here is that the learning enablers outlined in this section, presumably in conjunction with these other attention biases, allow infants quickly to learn about the various physically and psychologically causal properties of objects and their attributes.

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Attention to Parts

A first bias relates to the way that infants, from early in the first year of life, are adept at dissecting objects into their component parts and subsequently recombining those parts into wholes (e.g., Biederman, 1987, 1990; Cohen & Younger, 1983, 1986; Slater, Mattock, Brown, Burnham, & Young, 1991; Triesman, 1985, 1986; Younger & Cohen, 1983, 1986). This capacity is crucial in the construal of a three-dimensional world from a two-dimensional retinal image, and it allows object recognition even if components of an object are occluded from view. For instance, in the context of adult visual perception, Triesman (1986) argued that "the analysis of properties and parts precedes their synthesis" (p. 117), and Salapatek (1975), in a more developmental view, claimed that "the very early perception of two-dimensional stimuli must be regarded as the perception of parts rather than wholes" (p. 226). Indeed, there is evidence that even newborns process simple stimuli such as red and green vertical and diagonal colored stripes in terms of their components (color and orientation) rather than in terms of separate properties (color or orientation) (Slater et al., 1991).

Over the past 20 years, a number of researchers have examined the role of parts and part structure on infant, children, and adult object recognition and categorization. With regard to adults, Palmer, Rosch, and Chase (1981) found that they agree on the viewpoints of common objects that are more *canonical* than others, and that these viewpoints tended to reveal the more important parts in contour. Hoffman and Richards (1984) presented a principle whereby local minima are used to decompose contours into parts, while Biederman (1985) provided evidence that recognition uses inflection points—that is, the point in a curve where a change occurs from concave to convex, or vice versa—to determine a number of component parts. Moreover, parts that are engendered through these kinds of principles have been shown to have consequences in other tasks; for example, they are quickly identified as belonging to a particular form (Palmer, 1977), and they are considered to play an important role in the structural descriptions of objects (Hoffman & Richards, 1984). Likewise, 5-year-old children group exemplars from one category more readily when they share parts, and they detect missing parts more quickly when they are external and affect structure (e.g., legs) than when they are internal and do not affect structure or shape (Tversky, 1989).

With regard to infants, Younger and Cohen (1983, 1986; Younger, 1993) used the habituation procedure to show that by 4 months of age infants can detect individual parts of an object—often a schematic animal—and by 7 months they can detect correlations among such parts. However, it is not until 10 months of age that infants can extract correlations of attributes that are embedded within a category (Younger & Cohen, 1986). More recently, Rakison and Butterworth (1998a, 1998b) used the object manipulation, or sequential touching, paradigm to test whether 14- to 22-month-olds' superordinate- and basic-like categorization is affected by object parts such as legs and wheels. In this paradigm, infants are presented with eight items—usually four each from two categories—and their sequential touching behavior is measured; systematic successive touches to exemplars of each of the categories is taken as evidence of classification. In one experiment, Rakison and Butterworth (1998b, exp. 1) presented infants with different-part contrasts—for example, animals

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versus vehicles or furniture versus vehicles—as well as same-part contrasts; for example, animals versus insects or insects versus furniture. Results revealed that infants categorized different-part contrasts approximately 8 months before they categorized same-part contrasts.

In a second experiment, Rakison and Butterworth (1998b, exp. 2) used the same procedure with 14-, 18-, and 22-month-olds but made modifications to the exemplars to examine more closely the basis for early categorization. Each infant was presented with four conditions with animals and vehicles, the specific exemplars being the same as those used in the earlier experiment. The four conditions are illustrated in table 7.1. In one condition, infants were presented with normal animals and normal vehicles (control task); in a second condition, infants were tested with a contrast of animals without legs and vehicles without wheels (no parts task); in a third condition, they were given a contrast of animals and vehicles, all of which had both legs and wheels (matched parts task); in a final condition, called the across-category confound task, infants were presented with two animals and two vehicles that possessed wheels but no legs, as well as two animals and two vehicles that possessed legs but no wheels. Examples of these stimuli are presented in figure 7.1.

The strength of the across-category confound task is that infants have a "choice" to categorize exemplars by superordinate category membership—by grouping, for example, all the animals together—or by part relations; for example, grouping together two unmodified vehicles with two animals with wheels. Infants' sequential touching behavior across the four contrasts supported the notion that until at least 1½ years of age, parts are sufficient and perhaps necessary for categorization of superordinate-like domains. Infants in all three age groups categorized the unmodified animals as different from the unmodified vehicles, suggesting that they indeed do have knowledge about these domains. However, all three age groups failed to categorize the animals as different from the vehicle when the exemplars possessed legs and wheels (matched parts task) as well as when the exemplars possessed no differentiating large parts (no parts task). The results from the final condition provided even stronger evidence for the role of parts in early categorization: infants at 14 and 18 months grouped objects on the basis of parts relations—that is, "things with legs" and "things with wheels"—rather than superordinate category membership, and infants at 22 months failed to categorize on either basis.

These experiments lend support to the idea that parts act as the basis for early categorization. The ability to form categories at the superordinate level is commonly viewed as evidence of an understanding of the conceptual, or nonobvious, relations that exist between members of those domains (Mandler & Bauer, 1988; Mandler, Bauer, & McDonough, 1991), yet infants' behavior in these experiments seemed to be driven very much by perceptual attributes. It may well be that infants, until at least 22 months of age, do not understand that relations exist within the members of a specific category—for instance, that cats have whiskers—as well as across the members of broader category; for instance, that cats, dogs, and zebras are mammals (Rakison & Butterworth, 1998b; Rakison, 2000a, b). Moreover, infants' behavior in the tasks suggests that they may have formed categories on-line; that is, the items used to form a category may be dependent on those that are available and the attri-

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TABLE 7.1. Object Manipulation Tasks and Exemplars in Rakison and Butterworth (1998b, Exp. 2)

Task	Animals	Vehicles
	Control task	
Animals versus Vehicles	Cow Dog Goose Walrus	All-terrain vehicle (ATV) Train Bus Motorbike
	Matched parts task	
Animals with wheels and legs versus Vehicles with wheels and legs	Cow with wheels and legs Dog with wheels and legs Goose with wheels and legs Walrus with wheels and legs	ATV with wheels and legs Train with wheels and legs Bus with wheels and legs Motorbike with wheels and legs
	No parts task	
Animals without legs versus Vehicle without wheels	Cow without legs Dog without legs Goose without legs Walrus without legs	ATV without wheels Train without wheels Bus without wheels Motorbike without wheels
	Across-category confound task	
Animals, two with legs and two with wheels versus Vehicles, two with legs and two with wheels	Cow with wheels and without legs Dogs with wheels and without legs Goose	ATV with legs and without wheels Train with legs and without wheels Bus Motorbike
	or	
Object with wheels versus Object with legs	ATV with legs and without wheels Train with legs and without wheels Goose Walrus	Cow with wheels and without legs Dog with wheels and without legs Bus Motorbike

butes possessed by those stimuli (see also Jones & L. Smith, 1993; Oakes, Coppage, & Dingel, 1997; Quinn, Eimas, & Rosenkrantz, 1993).

To investigate whether object parts are also important in early categorization at the basic level, Leslie B. Cohen and I (Rakison & Cohen, 1999) presented 14- to 22-month-old infants with the same four tasks used by Rakison and Butterworth (1998b, exp. 2) with contrasts of cows and cars. A basic-level contrast of this type differs from a superordinate contrast of animals and vehicles in that within-category similarity is relatively high—that is, cows look alike whereas animals do not—though the level of between-category contrast is equivalent in each case; that is, both contrasts are of animals and vehicles. As a further measure of infants' knowledge about

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Figure 7.1. Examples of stimuli used by Rakison and Butterworth (1998b, exp. 2).

the motions of the objects represented by the stimuli, we also recorded the number of times that infants made "rolling" or "walking/jumping" actions.

Infants' sequential touching behavior across the four tasks revealed that parts are important in basic-like categorization but that their importance varies depending on the level of within- and between-category similarity. As expected from the experiments by Rakison and Butterworth (1998b), infants at 14, 18, and 22 months of age categorized unmodified cars as different from unmodified cows. Yet in the task in which the parts were removed from the cars and cows, the youngest age group failed to categorize the stimuli as different. Moreover, in the task in which parts were attached to the cars and cows—that is, when they all possessed legs and wheels—the

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14- and the 18-month-olds failed to categorize the two stimulus sets as different. The 22-month-olds, in contrast, categorized the cars and the cows as different on all three tasks. Somewhat surprisingly, on the across-category confound task, infants in all three age groups categorized objects not by parts but instead as cows and cars. Overall, these results imply that infants attend to object parts in categorization, but that the level of within-category dissimilarity influences this bias and that perceptual features other than large, functional ones may play a significant role in class membership decisions. For example, infants at 18 months of age categorized cars and cows without legs and wheels as different but categorized cows and cars with both legs and wheels as equivalent. This suggests that in the absence of parts like legs and wheels, 18-month-old infants are able to attend to other feature similarities to form categories, but in a context when stimuli possess legs as well as wheels, the same infants attend to these parts in category formation.

The analyses of infants' functional responses provided an interesting insight into the reason for infants' attention to large parts. Although the overall number of functional behaviors by infants was low, the 14-month-olds as well as the 18-month-olds tended to "roll" and "walk" the toy stimuli with parts more than those without parts, and the 22-month-olds made a similar number of functional responses to those objects with and without parts. Of particular interest was that infants in the across-category confound task, who did not categorize on the basis of parts, tended to "roll" or "walk" the novel stimuli on the basis of their parts rather than their ontological status. That is, instead of making a car with legs "roll," infants made it "walk," and instead of making a cow with wheels "walk," infants made it "roll."

In a further series of experiments, George Butterworth and I (Rakison & Butterworth, 1998a) examined whether infants attend to parts per se or to the structure given by those parts. Infants may group together animals and furniture not because they both possess legs but rather because those parts give objects a similar overall structure; namely, a body with four vertically oriented attributes in a rectangular configuration. In one experiment, for example, we again used the object-manipulation paradigm with 14-, 18-, and 22-month-olds to examine categorization of animals, vehicles, and furniture when object structure was modified in orientation alone (e.g., four legs inverted and placed on top of the body of an object) or in orientation and configuration (e.g., four legs at tangential angles). Infants' sequential touching behavior was consistent with the idea that part structure is as important, if not more important, than the presence of parts themselves; animals were categorized as different from vehicles by 14- and 18-month-olds when parts maintained their configuration but were placed in a novel orientation, but they were not categorized as different when the common configuration of parts was violated.

To summarize, the research on 1- to 2-year-old infants' behavior in the sequential-touching paradigm suggests that parts and part structure play a primary role in category membership decisions. The evidence suggests that object parts such as legs and wheels are sufficient and, depending on the levels of within- and between-category similarity, necessary for categorization. The data also suggest that infants between 14 and 18 months are flexible in the categories they form; that is, they appear to create categories on-line that are dependent on the available stimuli and the attributes of those stimuli. Finally, the data suggest that infants around

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14 months of age may have started to connect certain motion characteristics with objects, or more specifically, with the part that are causally related to those motion characteristics.

The Role of Movement

A wide range of evidence supports the idea that movement has an "attention-grabbing" effect on infants. From birth, infants are able visually to track a simple stimulus, and they are more likely to do so for facelike configurations than for random or blank stimuli (Morton & Johnson, 1991). Newborns also show a visual preference for moving stimuli over static ones (Slater, 1989), and infants between the ages of 7 and 21 weeks orient to a small moving object in a visual field of static objects (Dannemiller, 2000). By 2 to 3 months of age, infants' ability to perceive particular characteristics of objects is dependent on the presence of movement cues. For instance, 2-month-olds who are habituated to a moving rod that is partly occluded perceive it as a complete object (Johnson & Aslin, 1995), yet 3- to 4-month-olds do not perceive the rod as complete if the two parts fail to move (Kellman & Spelke, 1983).

Moreover, by 3 months of age infants prefer a moving point-light display of a human to an unstructured or random point-light display, and they show no preference between the same displays when they do not move (Bertenthal, 1993). Along the same lines, recent evidence from Arterberry and Bornstein (in press) suggests that 3-month-old infants can categorize point-light displays of animals as different from point-light displays of vehicles. According to Kellman (1993), this role of motion in specifying an object's characteristics is an innate or maturational *primitive process*, and to this end there is evidence that infants as well as adults more easily detect an object's properties when it moves than when it is still (Burnham & Day, 1979; Washburn, 1993; Werker, Cohen, Lloyd, Casasola, & Stager, 1998).

Why should motion have such a facilitating effect on infants' attention to object properties? I posit that infants, as with other species, find object motion particularly salient because of an evolved adaptive mechanism for predator and prey detection. Many species are known to have visual systems that are specifically designed for their particular environments and the tasks faced in those environments; for instance, the frog retina has a "bug" detector, and the rabbit retina is thought to have, among other things, a "hawk" detector (Marr, 1982). Perhaps more compelling, the seminal work of Seyfarth and Cheney (1986; see also Evans & Marler, 1995) suggests that young vervets use motion cues to identify and categorize potential predators, and they will often make mistaken "eagle" alarm calls for a variety of birds and even nonbirds such as swiftly falling leaves. If other species show such an innate sensitivity to motion, usually in the context of predator or prey identification, I see no reason why humans would not have evolved the same, or at least a similar, visual mechanism. Such a mechanism would be crucial in drawing infants' attention to specific, significant information, or, as Dannemiller (1998) put it, "selective attention mechanisms are necessary to resolve the problems of where to look next in a visual field that is populated with multiple objects (as it usually is)" (p. 269). For the

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time being, the important message here is that movement, presumably both at a global level (object moves from A to B) and at a local level (part X moves on object A), is highly salient to infants.

The Role of Size

A minor early attention bias concerns the fact that large parts and large objects are initially more salient than small parts and small objects. I use the terms *large* and *small* in relation to size to refer to relative rather than absolute terms. Thus, a large part can be defined as one that composes a relatively high proportion of the whole object of which it is a component, and a small part can be defined as one that composes a relatively low proportion of the whole object of which it is a component. It remains to be seen what the exact part-to-object ratio need be for a part to be considered large or small; nonetheless, it has been shown that, for adults at least, the salience of a part increases as the ratio of its observable area to the observable area of the whole silhouette increases. Research on the role of relative part size on infants' categorizing behavior has not yet been performed; however, there is evidence that newborns possess shape and size constancy (Slater & Morrison, 1985) and that when presented with two identical cubes, they look longest at the one that produces the larger retinal image (Slater, Mattock, & Brown, 1990).

Concept Formation and Associative Learning

Thus far I have argued that infants possess inherent mechanisms that bias them to attend to object parts, and the movement of large attributes and of objects. The section on the role of object parts outlined evidence that infants at the beginning of the second year categorize animate entities and inanimate objects on the basis of attributes such as legs and wheels. This behavior is clearly a case of what is known as perceptual categorization. But how does this early predilection to attend to object parts, as well as to movement, facilitate the development of what is known as "conceptual" categorization? That is, how can attention to the attributes of objects lead to the emergence of concepts that encapsulate motion properties such as self-propulsion and agency or psychological properties such as goal-directed action? This kind of knowledge is what, among other things, allows adults to make the distinction between animate entities and inanimate objects. It is important to note, however, that the categories "animate" and "inanimate" are not naturally occurring in the same way as, for example, "dogs" or "Rottweilers"; more probably, they are constructs of the adult mind that provide an additional, extremely broad grouping of things in the world.

The functional analysis provided by Rakison and Cohen (1999) is, in my opinion, a provocative clue to the nature of the development of infants' knowledge about different object kinds. Recall that the data suggest that infants at 14 and 18 months are starting to relate a movement pattern with the object part with which it is causally related, whereas infants at 22 months have associated movement patterns with

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specific parts as well as the body of objects. My claim is that the distinctive aspect of attributes such as legs and wheels is that they move, and, perhaps more important, this movement tends to cooccur with a function, motion, or psychologically related act. For example, the legs of an animate entity move predominantly when that entity engages in some kind of locomotion, the wheels of a vehicle move only when that vehicle itself is in motion, and the fingers of an empty hand move conjointly when about to engage in a goal-directed grasp. The salience of local motion and of global motion—that of the attribute and of the whole object, respectively—together are powerful attention-grabbing stimuli; however, it is that infants are able to associate these two dynamic cues that allows them ultimately to go beyond information given in the perceptual array.

My broader claim is that infants possess an inherent domain–general association learning mechanism that, following detection of dynamic as well as static attributes by one or more of the perceptual systems, encodes highly correlated relations among them. In the case of dynamic cues—those relating to motion, for example—these correlations connect a readily perceptually available attribute with a particular action. The result is a represented associative link that embodies what has come to be labeled "conceptual" knowledge or understanding. But I argue that there is no need to apply the kind of rich description to these representations that has been suggested in previous formulations of concept development (e.g., Leslie, 1995; Mandler, 1992; Spelke, Breinlinger, Macomber, & Jacobson, 1992). Instead, these associations should be considered in terms of an *expectation*, such that the perception of one constituent of the correlation elicits an anticipation about the presence of the other (for a more detailed and general discussion of infants' expectations see Haith, Wentworth, & Canfield, 1993; Roberts, 1998). For the kind of expectations outlined here, what is connected is a readily perceptually available attribute—wings, for instance—with an intermittently perceptually available motion characteristic; for example, flying, self-propulsion, and so on.

Initially, after having encountered only a few instances that exhibit a particular relation, the perception of one component of that relation will not necessarily trigger the other component; to depict the noncausal state of this preliminary connection, elsewhere I have labeled it an elementary association (Rakison & Poulin-Dubois, 2001, 2002). Following repeated exposures to the same correlation, the strength of the correlation becomes such that the perception of one part of the correlation can trigger the expectation that the other is present—what I have labeled a secondary association (for a similar view of gradual strengthening of representations, see Munakata, McClelland, Johnson, & Siegler, 1997). It is worth noting at this juncture that I use the term *association* to refer to the mental representation of correlations that exist between spatially or temporally co-occurring attributes objects, properties, or events in the world. I do not use the term to refer to any kind of a stimulus-response connection, because these invariably involve the self rather than external properties exclusively; in other words, I am discussing stimulus-stimulus relations.

I posit that the represented associations, once they have developed sufficiently to allow for expectations to be triggered, may be thought of as the preliminary development of a *meaning* about an object or its attributes. That is, objects that were initially grouped together because they share similar attributes would start to be categorized

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as equivalent because they engage in similar kinds of actions or functions. Over time, additional attributes and properties of an object become associated with the initial pairing—even if such attributes are not causally related to the action involved—and ultimately, with the onset of labeling, these associations are generalized to whole objects. Even at this point, however, the different kinds of information about objects embodied in representations need not be thought of as conceptual or nonobvious; rather, there is a connection between static and dynamic perceptual cues that allows for categorization that appears to go beyond the information given in the perceptual array.

Within this framework, I propose that any attributes or properties of objects, even those that are classically held to be nonobvious, can be thought of as perceptual because they are represented following input through one or more of the sensory systems. A good deal of this information, particularly that which is referred to as perceptual in the literature, is dealt with by the visual system, as when, for example, an infant observes that a dog has a tail and four legs (see also Eimas, 1994). As an infant becomes older and language becomes a viable input system, information about objects—particularly that referred to as nonobvious—will be acquired through labeling. The information that arrives through these distinct input systems need not be represented in fundamentally different ways. In a similar manner, it is assumed that perceptually readily available surface features—facial features, for example—are represented similarly to those surface features that are less perceptually accessible (e.g., the paws of a dog or a person's belly button). To put it plainly, there is no logical reason why "unobservable" properties should be representationally distinct, in a qualitative sense, from those that are more readily available in the perceptual input.

To make this process clear, consider the example of how infants might come to learn about the properties of dogs, and in particular the motion properties of dogs. This progression is illustrated in figure 7.2. Most probably, infants' first basis for categorization of dogs would be their larger parts such as legs, faces, or tails (e.g., Quinn & Eimas, 1996a; Rakison & Butterworth, 1998b). In line with the global-to-basic trend observed in a number of studies (e.g., Mandler et al., 1991; Mandler & Bauer, 1988; Younger & Fearing, 1999), attention to such parts would not allow for a division of, say, dogs from cats or other similar mammals, but it would allow for a division to be made between dogs and vehicles, plants, tools, and so on. At the same time, infants may start to discriminate the different motion patterns that are characteristic of different object kinds; for example, they might discriminate smooth from irregular motion or self-propelled from caused motion. They would not know, however, which object kinds perform which particular motion (Rakison & Poulin-Dubois, 2001). During the second year of life, following repeated observations of dogs engaging in particular motions (e.g., nonlinear movement, self-propulsion), I suggest that infants would start to associate legs with those motions; that is, "things with legs are self-propelled." Initially, after only a few observations, this association would be weak and would not trigger the kind of expectation discussed earlier; however, following a considerable number of encounters with instances that exhibit the same part-motion relation, the association would become strong enough to allow for prediction of one attribute based on the perception of the other. Again, infants at this point would not categorize dogs as different from cats, but they would treat

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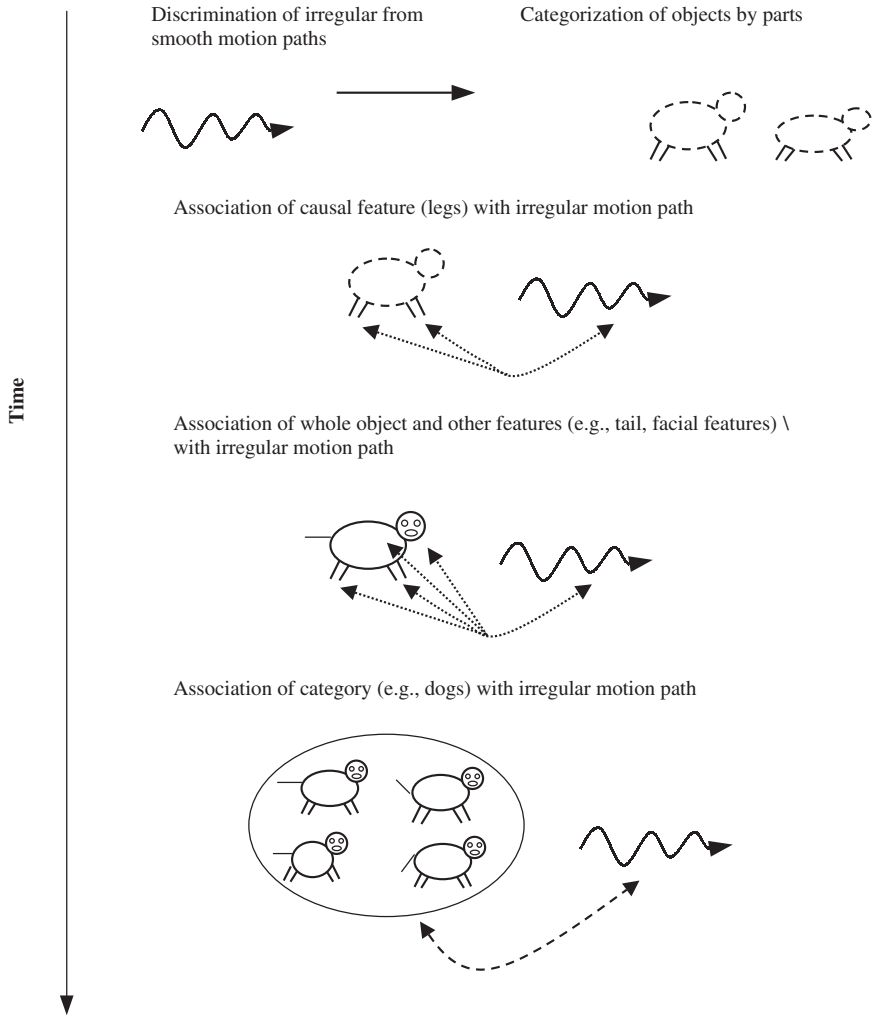


Figure 7.2. Depiction of the development of association-based representations. Time is given on the vertical axis.

furniture, whose legs do not afford movement, as different from dogs. Over time infants would detect other attributes of dogs, and these would start to be incorporated into the initial association, such that the representation can be thought of in terms of "things with legs, eyes, fur, and a tail walk." This process, I suggest, would occur in parallel for a number of different properties connected with dogs; thus, dog mouths would become associated with barking, and tail-wagging would become associated with alert, happy behavior (on the part of the dog, that is).

With the onset of language, I speculate that the associations are drawn together under one label, so that the utterance "dog" triggers the attributes, both static and

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dynamic, that have come to be associated with dogs. Indeed, I suggest that language plays a singularly important role in this process because it not only allows for the acquisition of properties that are nonobservable but also speeds up the process of representational development by allowing rapid one- or two-shot learning—a form of fast-mapping—of such information, which is then applied across the category. In other words, in contrast to information about objects that is encountered through the visual, tactile, and auditory systems, input from language is assumed by infants to refer not to one particular exemplar but rather to all members of the basic and superordinate category to which the original referent is drawn. For example, if an infant is told that a cat possesses a brain, or cat DNA, following only one or two such labeling episodes, this fact will be extended to dogs, people, cows, and indeed all those exemplars that are considered alike to a cat. This notion, though currently lacking direct empirical support, is in line with infants' overextension errors on the basis of perceptual or featural similarity (see e.g., Nelson, 1974; Rescorla, 1980), preschoolers' attribution errors (Carey, 1985), connectionist simulations of match-to-sample studies (chap. 14), and a number of constraints that are thought to be particular to early word acquisition (see e.g., Markman, 1991; chap. 11).

This proposed view has at its core the idea that representational development is a process of enrichment rather than a stagelike shift in conceptual format and structure. Through a continual accumulation of associations that includes both static and dynamic readily perceptually available properties, as well as those that are less readily perceptually available, infants progressively develop concepts for animates and inanimates that incorporate the various characteristics that make the members of those categories distinct (for adults at least). In this sense, this proposal is very much in line with the view of Quinn and Eimas (1997, 2000; Eimas, 1994; chap. 3). My view differs, however, in a number of fundamental respects from previous proposals that are also grounded in perceptual information. Most notably, in my opinion the current proposal is not weakened by the *insufficiency of constraints* argument that has been levied against previous similarity-based formulations of concept formation (Keil, 1981, 1991; Murphy & Medin, 1985). Recall that the claim is that theories of conceptual structure and content that are based on similarity—as is a theory grounded in the ability to detect and encode clusters of correlated attributes—are doomed to fail because there are so many similarities (or correlations) to which, in principle, one could attend that it is impossible to know which ones are important for category membership. I argue that the position outlined in this chapter overcomes this problem by highlighting attention biases that help to guide infants, as well as children and adults, toward particular attributes and relations among attributes. And it is no accident that these relations tend to be those that are causally relevant and that are good predictors of category membership; the human mind has evolved to carve nature at its joints in a parsimonious, productive, and generally veridical manner.

This view also differs from previous perceptually based formulations in that it outlines how the continuous addition of associative links can operate as a mechanism for representational change. According to previous perceptually based theoretical proposals of concept development (e.g., Eimas, 1994; Jones & Smith, 1993; Quinn and Eimas, 1997), it is not clear how infants develop knowledge that goes

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beyond the information available in the perceptual input. Eimas (1994), for example, writes that "the common aspects of the features for animate things, for example, biological motion, that are found in the categorical representations for different species are presumed to be recognized and abstracted" (p. 87). According to Eimas, this process of "abstraction" is a form of categorization that leads to the formation of associations between, for example, facial features and legs with locomotion. This is a reasonable claim; yet the continual addition of associative links does not in and of itself explain how infants' concepts start to incorporate that which is held to be nonobvious knowledge. In other words, it does not explain how quantitative representational enrichment leads to qualitative representational change or, at least, the appearance of qualitative representational change. I too argue that infants' representational development is best characterized as a process of enrichment, with associative links added over time. But in addition, I posit that an ever increasingly sensitive perceptual system, along with the emergence of language and advances in motor skills that allow for greater exploration of the environment, causes the appearance of qualitative change. To be more specific, my claim is that infants' representations appear to undergo conceptual change as different kinds of information becomes available to the sensory systems, but the basic learning process is the same for the acquisition of perceptual and "conceptual" information (see also Madole & Oakes, 1999; chap. 6). An infant cannot understand that a dog has a heart because it simply does not have access to this kind of information; however, once language becomes a viable input system, such details can be incorporated into the already existing representation. There is no reason why this process should be thought of in terms of qualitative change, as is often the case in the developmental literature (e.g., Gelman & Markman, 1986; Carey, 1991, 1995). Rather, it is just another case of perceptually given information becoming associated with that which was earlier acquired through the various sensory input systems.

One of the key claims for the qualitative representational difference between perceptual and so-called nonobvious properties is that preschoolers tend to choose the latter to make category membership decision. For instance, when 4-year-olds are asked which of two objects belong with or are "like" a target object, they tend to choose nonobvious properties such as "warm-blooded" or "has a heart" over perceptual features, such as shape, when the two are pitted against each other (e.g., Gelman & Markman, 1986, 1987). However, I believe that such data fit comfortably with the kind of perceptually grounded proposal presented here. After all, this pattern of behavior is to be expected if, as hypothesized, toddlers and young children treat labeled properties such as "this has a brain" as applying broadly across the category of the original stimulus as well as those categories that are judged to be similar to that first category. If this is the case, then children's behavior in the sort of inductive inference tasks described earlier should not be surprising, in that when given a choice between a highly (if not perfectly) correlated property for category membership and a less highly correlated property for category membership, children choose the former over the latter. For instance, it is true to that all dogs possess a heart but not all dogs have tails, and so if the two properties are pitted against each other, the property "has a heart" is a superior bet on which to make an inductive inference. It is important to note, however, that I am not saying that perceptual, surface proper-

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ties play no role in category membership decisions. On the contrary, I believe that such properties are primary in everyday judgments about "what something is." However, given a choice between two properties, infants, children, and adults will choose the one that is better correlated with category membership (for data supporting this view, see chap. 14).

The continuation of this idea is that in contrast to the widely accepted view that people employ causal explanations to make sense of correlations among features (e.g., Medin & Ortony, 1989; Murphy & Medin, 1985), I propose that it is the associations between properties and causal features formed during infancy that bind or structure concepts. Lay theories, I wish to argue, are later attached to these associations either during formal learning or as and when needed. In many cases, adults know that two features are correlated, but they may have no understanding of the causal relation between those two features, nor need they develop some kind of lay theory to explain it in order to possess a functional concept. For example, I know that the computer I am working on, as with all computers, has some kind of silicon chips inside it, but I have not attempted to develop an explanation for how the chips make the computer function, though I could develop such a theory on-line if one were needed for some reason. Similarly, Smith (2000; see also chap. 11) has argued that the model for all cognition is the dynamic creation of a moment of knowing out of previous moments of knowing; that is, perceiving and remembering are very much influenced by "the immediate input, the-just preceding events, and the long-term experience of the perceiver" (Smith; 2000, p. 96). This does not mean, however, that the view presented here is completely at odds with the idea that theories help to provide categories coherence. For instance, Murphy and Medin (1985) argued that "concepts that have their features connected by structure-function relationships or by causal schemata of one sort or another will be more coherent than those that do not" (p. 313). My claim is that the causal link between attributes is inherent in the association between them that is formed early on in life, and when lay theories are developed to explain this causal link they will act to bind or cohere further the concept.

On the surface, at least, it is counterintuitive to consider that gradual associative learning can lead to qualitative changes in behavior. It is therefore worth briefly considering current work on connectionist models of cognition that attest to such a finding (see chap. 14). In particular, Schafer and Mareschal (2001) recently endeavored to model the behavior of 8- and 14-month-old infants in label-object associative learning tasks by Stager and Werker (1997). The results of the original study revealed that the younger age group discriminated two highly similar phonemes (bih-dih) whereas the older age group could only discriminate two relatively distinct ones (lif-neem). To explain these results, Stager and Werker (1997) posited that older and younger infants process the phonemes in qualitatively different ways, such that the 8-month-olds processed the labels phonetically and the 14-month-olds processed the labels phonemically. Furthermore, Stager and Werker (1997) suggested that stagelike transitions were responsible for these qualitatively distinct types of processing. Schaffer and Mareschal (in press) tested this interpretation by examining the ability of "younger" and "older" networks to learn specific label-object pairs within an artificial language. (The "younger" networks were exposed to fewer ex-

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amples of the artificial language than the "older" networks.) Despite the fact that the connectionist models relied only on associative, continuous learning, the simulations matched closely the behavior of the two age groups in the study by Stager and Werker (1997); that is, the older networks discriminated only the dissimilar phonemes, and the younger network discriminated the similar phonemes.

Can Infants Form Associations?

What evidence is there that infants are capable of forming association-based representations? There is now a considerable database that reveals that well within the first year of life, infants develop the ability to detect and encode clusters of correlated attributes. For instance, by 7 months of age infants perceive correlations among the attributes of a schematic animal, and by 10 months they can extract the same correlations when embedded in a category (Younger & Cohen, 1986; for a review see chap. 4). During the same period, infants also start to attend to dynamic relations. By 4 months, they are capable of detecting nonarbitrary relations, such as the relationship between the shape of a mouth and a vowel sound (Kuhl & Meltzoff, 1982, 1988), and by 7 months they are sensitive to arbitrary intermodal relations, such as the relationship between temporally synchronous vocalizations and moving objects (Gogate & Bahrick, 1998). Notably, however, the ability to process dynamic relations in the first year of life appears to occur only for those that have been frequently encountered by the infant—as is the case for mouth opening and vowel sounds, or in the presence of a *facilitating cue*, such as, in the case of a sound and object relation, temporal synchrony, intensity shift, or common rhythm. This suggests that although infants' associative learning mechanism allows them to learn relations among static and dynamic stimuli in the first 12 months of life, it is not yet fully operational.

Between 12 and 14 months, infants become considerably more adept at detecting and encoding correlations among auditory and visual dynamic cues. An excellent demonstration of this developing ability can be found in a study by Werker et al. (1998) in which 10- and 14-month-old infants were tested with the *Switch* design in the habituation procedure. In the habituation phase of an experiment with this design, infants are shown attribute A^1 paired with attribute B^1 in one trial and attribute A^2 paired with attribute B^2 in another trial. Once habituated to these stimuli, infants are presented with a trial in which the attributes pairings are switched—for example, attribute A^1 is shown with attribute B^2—and a trial in which the attribute pairings remain unchanged from that shown during habituation. In the work by Werker et al. (1998), a label ("lif" or "neem") was one attribute in each habituation trial, and an object (a dog or a truck) was the other attribute. This design is especially ideal for examining infants' ability to learn correlations, because relatively long looking time to the "switch" test trial can only be caused by infants' detection of a novel attribute pairing and not by the introduction of a novel attribute (because no novel attributes are introduced). Indeed, Werker et al. (1998) found that 14-month-olds, but not younger infants, associate a label with an object and do so only when the object in question is moving during the habituation trials. Although infants younger than

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14 months did not learn the relation between a label and a word, additional experiments by the same authors (Werker et al., 1998) revealed that they did process independently each of the attributes the events; that is, they processed the object and the label but did not associate them.

There is also evidence that infants between 14 and 18 months are starting to develop the ability to learn correlations that exist between two perceptual cues. Madole, Oakes, and Cohen (1993), for example, used the object-examination paradigm with 10-, 14-, and 18-month-olds to test whether infants could associate an object's form—that is, its shape—with its function. During the habituation phase of the task, infants were familiarized with two different objects that had the same form–function relation (e.g., objects with wheels that rolled). They were then tested with an object that maintained the familiar form-function relation, an object that violated the form-function relationship, and a completely novel object. Results revealed that the 18-month-olds, but not the two younger age groups, learned the correlation between form and function. These data were extended by Madole and Cohen (1995), who examined 14- and 18-month-old infants' ability to attend to form-function relations akin to those found in the real word. Infants were familiarized with objects in which a particular part (e.g., wheels) was correlated with a particular function (e.g., rolling). Infants at both 14 and 18 months learned the relationship between parts and their function, as suggested by their relatively long attention to a stimulus that violated the learned correlation. The authors also familiarized infants with part-function correlations unlike those found in the real world; that is, the form of one part predicted the function of another part. Under these conditions, the younger infants learned the form-function correlation, suggesting that they are unconstrained in the correlations to which they will attend, whereas the older infants responded as if they have not learned the correlation.

As the studies presented in this section suggest, there is considerable evidence that infants in the first 1½ years of life begin to attend to correlations not only among static attributes but also among dynamic ones. Although this evidence suggests that the mechanism for learning about animate entities and inanimate objects comes on-line early in life, it does not provide direct confirmation that infants are able to represent these things via the relation between moving parts and motion characteristics. To examine precisely this issue, I have recently examined infants' ability to attend to correlations among features in a motion event (Rakison & Poulin-Dubois, 2002). In particular, I used a variation of the switch design to test the idea that infants are biased to attend to dynamic attributes in an event and to correlate those attributes. In one series of studies, 10-, 14-, and 18-month-old infants were habituated to two events in which an object moved across a screen. Each object consisted of a combination of a unique moving part (yellow cigar shapes that moved horizontally or green diamond shapes that moved vertically), with a unique body (red oval or blue pot-shape) and a unique motion path (rectilinear or curvilinear). Infants might see during habituation a blue-bodied object with yellow horizontally moving parts traveling along a linear trajectory and a red-bodied object with green vertically moving parts traveling along a curvilinear trajectory (the combination of parts, body, and motion path were counterbalanced across the participants). Examples of the objects and the motion paths are shown in figure 7.3.

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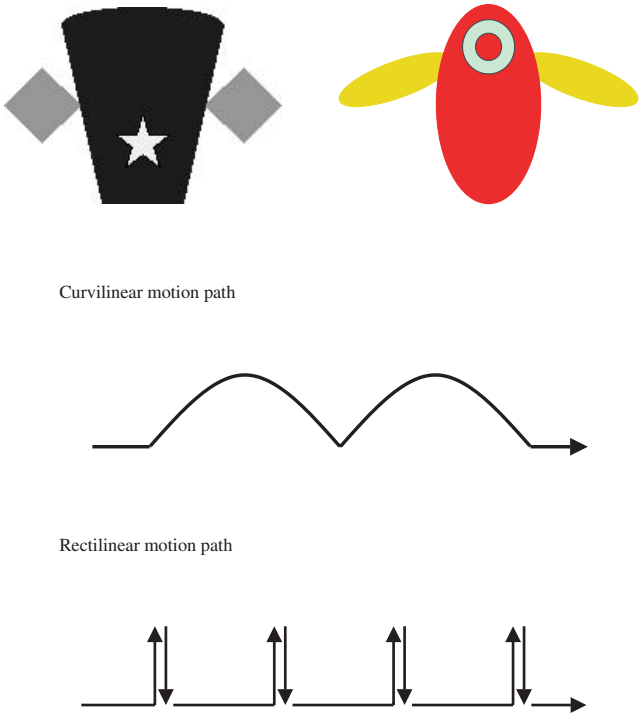


Figure 7.3. Examples of objects (parts and body) and motion paths of stimuli used by Rakison and Poulin-Dubois (2002).

During the test phase of the experiment, infants were shown one trial in which the parts were switched in relation to the pairing seen during habituation, one trial in which the body was switched in relation to the pairing seen during habituation, one trial in which the motion was switched compared to that seen earlier, and one trial that was the same as that seen during habituation. For example, given the two habituation stimuli described, in the *parts switch* infants might see the blue-bodied object traveling linearly but with green horizontally moving parts, in the *object switch* they might see the blue-bodied object with green vertically moving parts traveling along a curvilinear trajectory, and in the *motion switch* they might see blue-bodied object with green vertically moving parts traveling along a linear trajectory. It is important to note that during all the test trials, the object parts moved exactly as they had during habituation; in other words, the yellow parts always moved up and down, and the green parts always moved in and out. What is appealing about this design is that it presents infants with multiple correlations among the stimuli—namely, part-trajectory, object-trajectory, part-object, part-object-trajectory—and gives infants the "choice" to attend to one, none, or a number of them. If, during habituation, infants associated object parts with a motion path, they should look longer at the parts switch and the motion switch test trials than at the familiar test trial. If, on the other hand, infants associated the parts and body of the objects, they

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should look longer at the part switch and the body switch test trials than the familiar trial. If infants attended to the relation between object's bodies and motion trajectory, they should increase visual attention to the body switch and the motion switch test trials compared to the familiar test trial. Finally, if infants associated multiple attributes in the events, they should recover visual fixation to all three test trials relative to the familiar trial.

The results of the experiments not only were consistent with previous research on early attention to dynamic correlations (e.g., Madole & Cohen, 1995; Werker et al., 1998) but also supported the idea that correlations between dynamic cues are privileged in the infants' processing of events. Infants at 10 months of age failed to learn any of the correlations, and a subsequent experiment revealed that they could process independently the static bodies of the objects but not the dynamic parts or motion trajectory of the object. In contrast, infants at 14 months attended to the correlation between an object's parts and its motion trajectory. This is not to say, however, that 14-month-olds did not attend in some respect to part-body relations. As can be seen in figure 7.4, the 14-month-old infants did not recover visual attention to the motion switch, as might be expected. This finding was attributed to the fact that many infants (15 out of 24) looked away from the motion switch test trial early in the event because the part-body relation was not violated and therefore did not see the violation in the part-motion relation. As can also be seen in figure 7.4, the 18-month-olds attended to all of the correlations among the attributes in the event; that is, they had learned that objects with particular parts and a particular body moved in a certain way.

Recall that I earlier argued that the motion of functional parts is crucial in drawing infants' attention to causally relevant relations involving those parts. To test this idea, in a separate experiment 14- and 18-month-olds were presented with the same habituation and tests events as those described here, but in these events the parts of the objects did not move. As can be seen in figure 7.5, 14-month-old infants under these conditions no longer learned the correlation between object parts and a motion trajectory (or any other correlations in the events), and 18-month-olds attended only to the correlation between parts and motion trajectory. The behavior of the older group in this experiment suggested to me that by this age they had learned—that is, they expected—that the external parts of an object are often related to the motion characteristics of that object. Although admittedly speculative based on this limited evidence, this conclusion is consistent with the "hopping" and "rolling" behavior performed by infants on the stimuli without moving parts in the study by Rakison and Cohen (1999).

In line with the position presented earlier in the chapter, these data provide strong support for the idea that infants start to learn about the "nonobvious" characteristics of objects, at least those related to motion, by forming associations between functional, causally relevant object parts and those characteristics. In a short period of time, presumably following repeated exposures to exemplars that exhibit similar relations, infants generalize this association to encompass whole objects. Concurrently, the strength of the associations develop such that expectations about the presence of one component is triggered by observation of the other component; in this way, infants come to expect that "things with legs walk" and "things with wheels roll," and they do not need to see such a thing engaging in an action to know that it is capable of that action.

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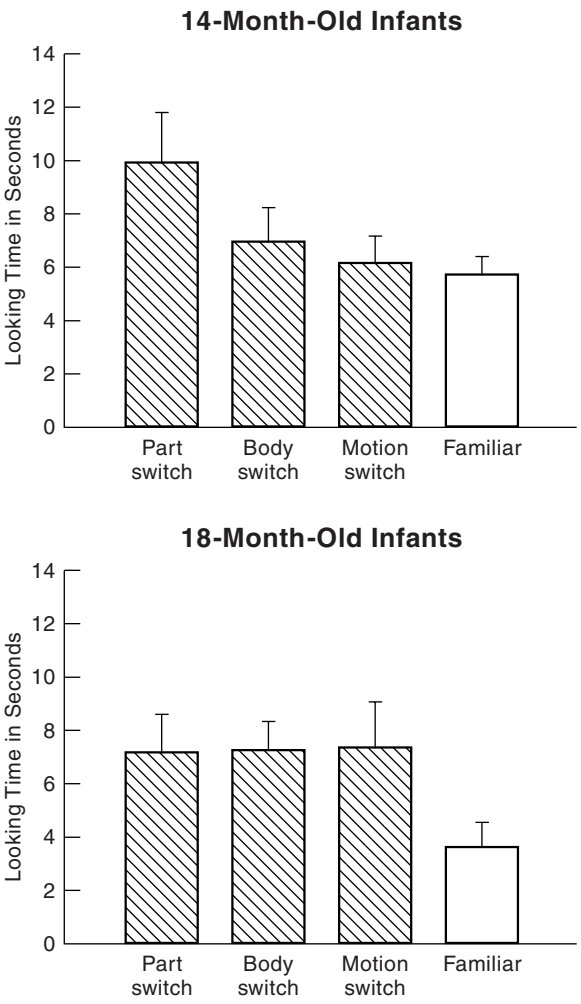


Figure 7.4. Infants' looking time (in seconds) to test stimuli with moving parts in Rakison and Poulin-Dubois (2002, exp. 1). The 14-month-olds looked significantly longer at the part switch test trial (in which the parts–motion correlation is violated) than at the familiar test trial (there were no other reliable effects). The 18-month-olds looked significantly longer at the part, body, and motion switch than at the familiar test trial.

Infants' Knowledge of the Properties of Animates and Inanimates

It is hopefully evident at this point that the claim here is that the formation of associations leads to a coherent concept rather than the other way round. According to some researchers (e.g., Mandler, 1992, chap. 5), it is an early concept of animal that underlies the formation of associations between attributes of that category and the

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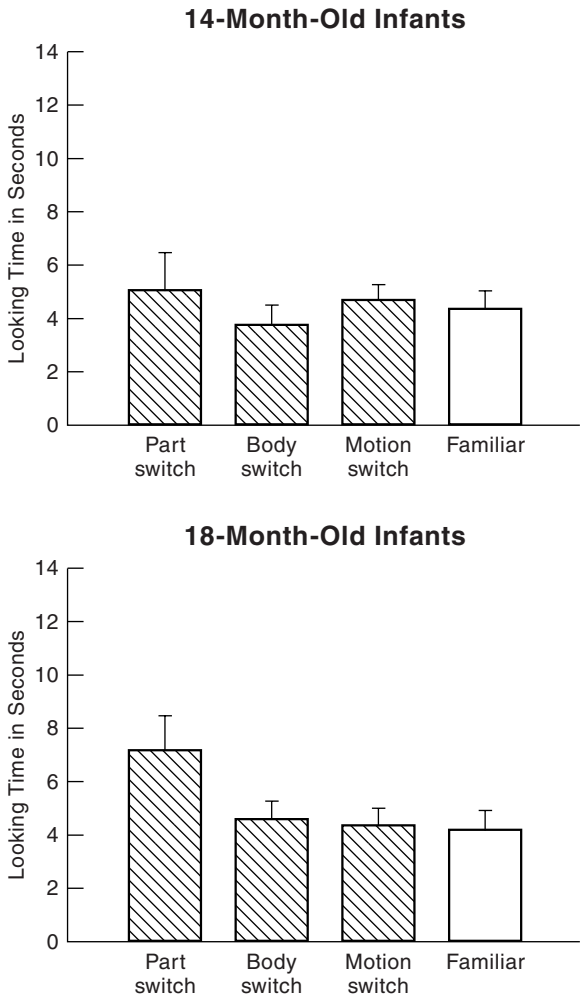


Figure 7.5. Infants' looking time (in seconds) to test stimuli with nonmoving parts in Rakison and Poulin-Dubois (2002, exp. 2). The 14-month-olds looked equally long at all four test trials. The 18-month-olds looked significantly longer at the part switch test trial (in which the parts–motion correlation is violated) than at the familiar test trial (there were no other reliable effects).

characteristics of those attributes (e.g., animals drink and sleep). In my view, this is a case of putting the cart before the horse. It seems the wrong way round to ascribe young infants with relatively advanced concepts that embody the properties of animates and the properties of inanimates and then to suggest that additional, presumably more concrete characteristics of objects are later added to these concepts. Is it not more parsimonious, and indeed fitting with the data, to presume that category and concept formation begins with rudimentary perceptually grounded concepts

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that are gradually augmented by the addition of information about static and dynamic properties as and when the sensory systems allow? Indeed, in a recent review of the literature on this issue, Rakison and Poulin-Dubois (2001) found very little evidence that infants under 16 months of age had associated motion- or psychological-related characteristics to animates or inanimates. The one exception to this rule was that in many cases infants had associated animate-appropriate properties to people (e.g., Poulin-Dubois, Lepage, & Ferland, 1996; Woodward, 1998, 1999).

If, as has been suggested (e.g., Mandler, 2000, chap. 5), infants have concepts for animates and inanimates—or perhaps more simply for animals and vehicles—one would expect them to display knowledge of the properties of the members of these categories. Accordingly, Mandler and McDonough (1996, 1998) have used the generalized imitation, or *inductive generalization,* procedure in a series of studies to test exactly this issue. Mandler (chap. 5) provides an excellent summary of these studies, and I highlight here only two aspects of the results that appear to provide support for a precocious infant concept former. First, when an experimenter models a dog drinking from a cup or a car being "started" with a key, 14-month-olds are more likely to generalize those behaviors to a novel appropriate exemplar (a cat for the drinking and a truck for the keying) than to a novel inappropriate exemplar (Mandler & McDonough, 1996). Second, when an experimenter models the same kind of events as those described earlier with both an appropriate and an inappropriate exemplar—for instance, a dog and a car are shown to drink from a cup—infants model the event with the correct novel category member. These results, in conjunction with those from a number of other experiments (e.g., Mandler & McDonough, 1998), were interpreted to mean that infants know that animals engage in certain actions and vehicles engage in different actions.

There are a number of reasons to doubt this rich explanation of the data, however. Elsewhere, I have argued that infants' behavior in inductive generalization studies can be explained by processes of perceptual matching and imitation (Rakison & Poulin-Dubois, 2000, 2001); that is, during the test phase of the task, infants do not demonstrate any kind of conceptual knowledge but rather they choose to imitate the observed events with the exemplar that is most perceptually similar to that used by the experimenter. Infants observe an action modeled by an experimenter with a certain exemplar—usually a prototypical animal or vehicle—and they then choose the most similar available stimulus to that exemplar during the test phase of the experiment. Indeed, in tasks in which the two test stimuli are perceptually similar—a cat and a dog, for example—infants tend to use both of them to repeat the observed action (Mandler & McDonough, 1998; chap. 5). There is no need for infants to make any kind of induction about the capabilities of either the exemplar used to model the events or the stimuli presented during the test phase. In all likelihood, adults would use the same kind of procedure if they were shown a completely novel task and offered the chance to repeat that task with one of two novel exemplars, one of which was similar to the original stimulus and one that was not similar.

I have recently used the inductive generalization technique to test 14- and 18-month-old infants' understanding of the motions of animals and vehicles (Rakison & Poulin-Dubois, 2000). Following a baseline phase, an experimenter modeled with

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a dog a motion event typical of an animal—jumping over a block or going upstairs—or modeled with a car a motion event typical of a vehicle: jumping from one ramp to another (what we called the Evel Knievil jump) or moving back and forth on a U-shaped ramp. To make the events salient to the infants, each one was accompanied by a neutral sound such as "wee" for the ramp-jumping event. These events are illustrated in figure 7.6. In the test phase, infants were presented with two stimuli, one from the vehicle domain (a motorbike, a truck, a bus, or a tractor) and one from the animal domain (a rabbit, a cat, a horse, or a cow). Infants' behavior in the task suggested that they have started to develop knowledge about the motion capabilities of animals and vehicles; that is, both age groups performed more actions with the appropriate exemplar in the generalization phase than in the baseline phase. However, the low level of responses in the task—both age groups repeated about one of the four events—and the fact that the 18-month-olds showed no improvement in performance over the 14-month-olds suggested that other factors may have been at work.

To test the robustness of these data and the hypothesis that perceptual matching and imitation drove infants' behavior, in a second experiment the same tasks were presented to infants except that an inappropriate exemplar was used to model the tasks; for example, the experimenter modeled the stairs event with a car rather than a dog. As can be seen in figure 7.7, a quite different pattern of results from that of the earlier experiment was found. Infants in both age groups chose to model the events with an inappropriate exemplar—using the "distractor" truck to go upstairs, for example—with the same regularity that they had used the appropriate exemplar—the "target"—in the first experiment. Note that although the performance of infants with the target stimulus appears to be high in the test phase,

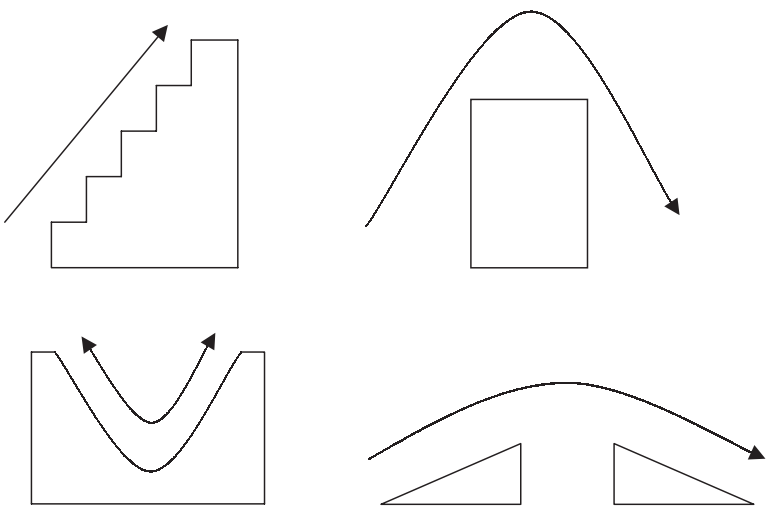


Figure 7.6. Schematics of motion events shown to 14- and 18-month-old infants in Rakison and Poulin-Dubois (2000).

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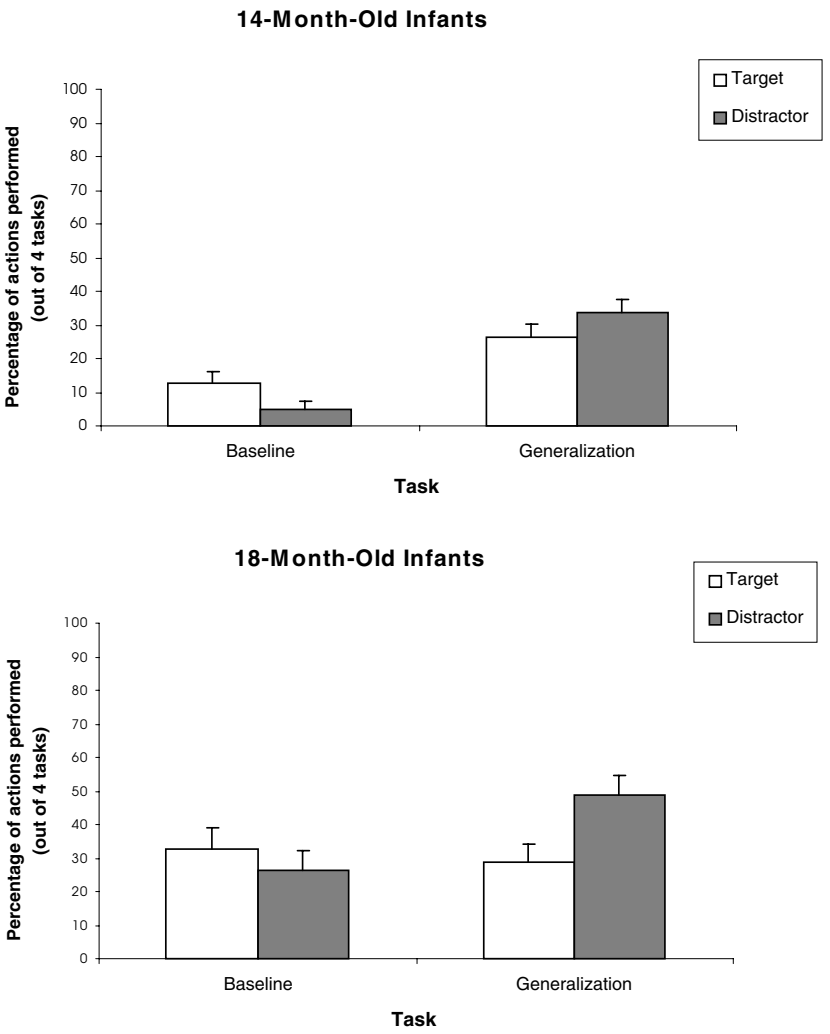


Figure 7.7. Percentage of 14- and 18-month-old infants in the inductive generalization paradigm using the target (appropriate) or distractor (inappropriate) exemplar across four tasks in which a distracter exemplar was used to model the action.

there was no significant increase between performance in that phase and that in the baseline condition.

It is possible that infants' tendency to model events with the distractor exemplar in the experiment just described resulted from the fact that they possessed little knowledge about the relatively specific events that were used. It could be argued, for example, that infants have had relatively little experience with events such as "going upstairs" or "a U-shaped ramp" and that in the absence of such knowledge,

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infants applied a matching and imitation procedure to complete the tasks. To address this issue, in a separate study (Rakison & Poulin-Dubois, 2000) infants were tested with general motion characteristics thought to be among the earliest learned (see e.g., Leslie & Keeble, 1987; Mandler, 1992), namely, the motion path of an object (smooth vs. irregular) and causal role (agent vs. recipient). To test early knowledge of path of motion, during the modeling phase for one condition infants observed a car rolling along a block, and for another condition they observed a dog "hopping" along a block. To test early knowledge of causal role, during the modeling phase for two conditions infants observed a dog move down a set of stairs and contact a car, which then proceeded to move. For one of these trials, infants were then asked to replace the agent with either a novel animal or novel vehicle, and in the other trial they were asked to replace the recipient with either a novel animal or novel vehicle. Note that although vehicles can appear to act as agents of an action and animals can be the recipients of an action, the prototypical roles played by agents and recipients tend to be animates and inanimates respectively.

Two separate sets of 14-month-olds were tested: One group observed the events modeled with the appropriate, or target, exemplar (as described earlier), and the other group observed the events modeled with the inappropriate, or distractor, exemplar; for example, a dog was observed hopping or a car was observed acting as an agent. Consistent with the results of the earlier experiments, infants repeated the events with the most similar exemplar to that observed during the modeling phase. Thus, infants would use, for example, a rabbit as an agent if the dog was observed in that role during the modeling phase. On other hand, and as can be seen in figure 7.8, infants would use a tractor as an agent if the car was observed as the agent during the modeling phase.

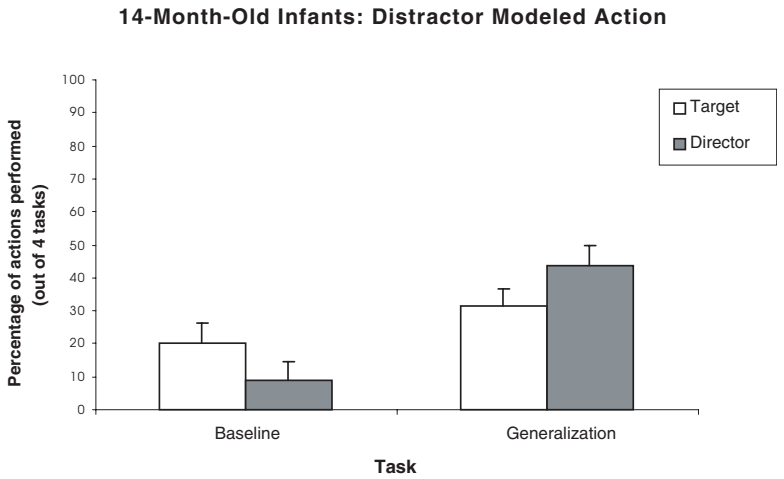


Figure 7.8. Percentage of 14-month-old infants in the inductive generalization paradigm using the target (appropriate) or distracter (inappropriate) exemplar across four tasks in which an inappropriate exemplar was used to model the action.

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I believe that these experiments, which constitute the first systematic attempt to examine infants' knowledge about the various motions of animals and vehicles, provide strong evidence that responses in the inductive generalization tasks are only partially driven by knowledge about the objects and events tested (see Mandler, 2000, chap. 5; Mandler & McDonough, 1996, 1998). Does this mean that infants around 14 to 18 months of age have not yet associated any motion characteristics with specific object kinds or the parts of those kinds? The habituation-based experiments described earlier suggest that it is not until 14 months or so that infants become able to represent dynamic correlations, and the fact that this ability emerges between 10 and 14 months implies that it is not until some time later that dynamic correlations in the real world are encoded. This is not to say, however, that infants necessarily brought no previous understanding about the motions of animals and vehicles to the inductive generalization tasks. For example, Mandler and McDonough (1998) found that infants in the inductive generalization paradigm, irrespective of the exemplar used in the modeling phase, will use a vehicle and an animal to repeat domain-general properties such as going into a building. It is quite possible that this sort of behavior suggests that infants possess broad knowledge about the activities of objects—that is, they are aware that things go in and out of different places—although it is just as likely that infants had no knowledge about these actions and used the available stimuli randomly. Nevertheless, the data presented in this chapter suggest that infants' responses in the inductive generalization task might well be guided by a process of perceptual matching and imitation rather than by a process of categorical induction. It is difficult to envisage an alternative explanation for why infants tended to imitate motion events with the inappropriate category after having seen an experimenter model those events with a prototypical member of that inappropriate category.

Concluding Remarks

I have argued in this chapter that the basic process underlying concept formation is that of associative learning. This inherent learning mechanism comes on-line in the first year of life, but it is constrained at this point by the relative incompetence of the various sensory systems. As infants' visual system improves, for example, so they are able to detect more fine-grained detail among the static features of objects, as well as being able to locate various dynamic cues at the local and global level. Likewise, as language starts to develop more fully in the second year, infants can use it to obtain substantial information about the properties of objects, most notably those that are not directly available in the perceptual array.

Infants' first concepts of animate entities and inanimate objects, and the category membership judgments that follow from them, are presumably grounded in static perceptual cues such as legs, wheels, facial features, and so on (Quinn & Eimas, 1996a, 1997; Rakison & Butterworth, 1998b). As dynamic information becomes available to the visual and auditory systems around the beginning of the second year of life, it is associated to the initial concepts such that infants begin to expect that an object that possesses a particular attribute will be capable of exhibiting a particular

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behavior, or that an object that exhibits a particular behavior will possess a certain attribute. This process gives rise to the appearance of qualitative change—in the sense that categorization might at this point be based on cues that are not readily available in the perceptual input—but it is clearly a case of gradual enrichment, or quantitative development. With the onset of language, concepts again seem to undergo a radical change. Objects are categorized on the basis of labels and shape, and then later, around 3 or 4 years, on the basis of nonobservable attributes depending on the context. As I have argued in this chapter, a relatively simple process of associative learning can account for these putative changes without the need to invoke some kind of specialized process for the abstraction of different properties of objects or dual processes that deal with perceptual and conceptual information independently (see Mandler, 1992, 2000).

The idea that not all correlations are born equal is likely to raise a few eyebrows. It has been widely accepted that the "insufficiency of constraints" argument is a fatal one for perceptually based views of concept development, largely because no one has yet proposed any kind of organizing principle that would act to constrain perception in the process of category and concept formation (Keil, 1991; Murphy & Medin, 1985). There is no longer a reason for researchers to be limited by such a view. As I have attempted to demonstrate here, it is important to discover on what properties or attributes infants, children, and adults base their category membership decisions and to seek to understand why those bases are salient or important rather than any others. The data reviewed in this chapter suggest that young infants are predisposed to attend to static cues and that older infants attend to dynamic cues—both at the global level, in terms of objects, and at the local level, in terms of the parts of those objects—and that by 14 months or so, they are able to associate these kinds of cues. I believe that this chapter provides a first step in providing a truly perceptual account of concept development by highlighting attributes that are potential candidates for attracting infants' attention toward the information that is relevant for category membership. Undoubtedly, there are many more important salient attributes of objects, as well as other mechanisms for learning—such as, for example, the predilection to habituate to familiar stimuli—that are not discussed here. One task for researchers in the years to come is to identify what these salient attributes and learning mechanisms are and to discover their role in the early representational development of infants.

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Notes

1. Throughout the chapter I use the terms *property* and *characteristic* to refer to any predicate that can be asserted of some or all of the members of a category (Hampton & Dubois, 1993). I use the terms *attribute* and *feature* to refer to particular aspects of a property or charac-

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teristic, whereby an attribute or feature is a property with a number of independent states or possibilities. For example, the property "is large" can be considered in terms of the attribute "size," and, likewise, the characteristic "self-propelled movement" can be considered in terms of the attribute "manner of initial motion."

2. I rely on Smith's (1995) definitions in my discussion of categories and concepts. To categorize something is to decide that it is a member of a category of objects, events, or properties that belong together. Mental representations of categories are used to decide which things belong together, and it is these mental representation that are referred to as concepts. Thus, in line with Margolis's (1994) view, the term *category* tends to refer to collections of things in the world, whereas the term *concept* refers to the representation of those collections.

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